

Dual-Axis Solar Tracker

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List of Principal Symbols and Acronyms

Acronyms

- **ADC:** Analog-to-Digital Converter
- **CAD:** Computer-Aided Design
- **DC:** Direct Current
- **GPIO:** General Purpose Input/Output
- **IoT:** Internet of Things
- **LDR:** Light Dependent Resistor
- **PV:** Photovoltaic
- **SoC:** System on Chip

Symbols

- **V:** Volt (Unit of electrical potential)
- **W:** Watt (Unit of power)
- **Ω :** Ohm (Unit of electrical resistance)
- **k Ω :** Kilo-ohm (10^3 Ohms)

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Abstract

The primary objective of this project is to design and implement an autonomous dual-axis solar tracking system that heavily focuses on power maximization and overall energy harvesting efficiency. Static solar installations suffer from inherent efficiency losses due to the changing incidence angle of sunlight throughout the day and across seasons. This project mitigates these losses by implementing a responsive, dual-axis tracking mechanism using an ESP32 microcontroller. Four Light Dependent Resistors (LDRs) are configured in a cardinal bridge to detect differential irradiance. The ESP32 processes these analog signals to determine the sun's relative position and actuates stepper motors via ULN2003A drivers to adjust the azimuth and elevation angles. By achieving near-perpendicular solar incidence continuously, the system records a vast increase in efficiency. Future iterations of this project intend to integrate Internet of Things (IoT) capabilities to allow for remote cloud-based telemetry and diagnostics.

Chapter 1

Introduction

1.1 Background Overview

The global demand for sustainable and renewable energy has accelerated the deployment of photovoltaic (PV) systems. However, static solar installations suffer from inherent efficiency losses due to the changing incidence angle of sunlight throughout the day and across seasons. To extract the maximum possible power from a solar panel, the sun's rays must ideally fall perpendicular to the panel's surface. When the angle of incidence deviates from this optimum, the energy yield drops significantly. This physical limitation creates the necessity for automated tracking mechanisms that can dynamically adjust the orientation of the solar panels to follow the sun's trajectory, thereby ensuring continuous optimization of energy harvesting efficiency.

1.2 Review of Previous Works

Early solar energy harvesting infrastructure primarily relied on fixed-tilt configurations. These systems were mathematically optimized for a specific geographic latitude but fundamentally failed to capture maximum solar irradiance as the sun's position shifted dynamically throughout the day. To address this, single-axis solar trackers were introduced, which successfully tracked the sun's east-to-west daily movement. While single-axis systems provided a notable increase in efficiency over static mounts, they could not compensate for the seasonal variations in the sun's elevation angle. Recent advancements in embedded systems have highlighted the superiority of dual-axis tracking mechanisms, which simultaneously adjust for both azimuth and elevation to maximize total power output.

1.3 Objective and Motivation

The primary objective of this project is to design and implement an autonomous dual-axis solar tracking system tailored specifically for power maximization. The core motivation behind this work is to completely mitigate the substantial energy losses inherent in traditional static solar setups. By utilizing a closed-loop control algorithm, the system maintains optimal solar panel alignment based on real-time light intensity gradients, ensuring a continuous and significant increase in efficiency.

1.4 Scope of Present Work

The scope of this project encompasses the hardware integration, mechanical actuation, and software control logic required to build a highly responsive tracking system. The central processing unit is an ESP32 microcontroller, which handles all sensory data and logic execution. For environmental input, four Light Dependent Resistors (LDRs) are configured in a cardinal bridge. The microcontroller actuates two stepper motors (driven by ULN2003A modules) to achieve near-perpendicular solar incidence, significantly boosting the panel's active efficiency. While current implementation focuses exclusively on mechanical alignment and power yield, the underlying architecture provides a robust foundation for future IoT integration.

Chapter 2

Hardware Components and Mechanical Architecture

2.1 Primary Hardware Components

The physical realization of the dual-axis solar tracker relies on the following core components to handle processing, sensing, and actuation:

- **ESP32 Microcontroller:** The main processing unit utilized to run the closed-loop control algorithm and actuate the motors.
- **$4 \times$ Light Dependent Resistors (LDRs):** Optical sensors arranged in a cardinal bridge to detect differential irradiance.
- **$8 \times 4.7 \text{ k}\Omega$ Resistors:** Arranged in four parallel branches, with two resistors connected in series within each branch (providing a $\sim 9.4 \text{ k}\Omega$ equivalent resistance to form the voltage dividers for the LDR array).
- **$2 \times 28\text{BYJ-48}$ Stepper Motors:** Actuators used for precise physical alignment. One handles horizontal (azimuth) movement, and the other handles vertical (elevation) movement.
- **$2 \times \text{ULN2003A}$ Motor Drivers:** Modules utilized to safely drive the stepper motors via digital step sequences.
- **Photovoltaic (PV) Solar Panel:** The primary panel mounted on the tracker to harvest energy.

- **5V-1A DC Power Adapter:** The primary power source providing sufficient current for the microcontroller and motor drivers.
- **Veroboard:** Used as the permanent prototyping foundation for circuit integration.

2.2 ESP32 Microcontroller

The central processing unit of the tracking system is the ESP32 microcontroller. It features a dual-core Tensilica Xtensa LX6 processor that provides the necessary computational power to execute the system's closed-loop control algorithm. The ESP32 is responsible for accurately processing the analog signals received from the sensor array to determine the relative position of the sun and drive the responsive tracking array.

2.3 LDR Sensors and Voltage Divider Configuration

To detect the sun's position throughout the day, the system utilizes four Light Dependent Resistors (LDRs). These optical sensors are strategically configured in a cardinal bridge (top, bottom, left, and right) to measure light intensity across multiple axes. To enable the ESP32's Analog-to-Digital Converter (ADC) to accurately read these varying light levels, each LDR is integrated into a voltage divider circuit. This is achieved using the eight 4.7 k Ω resistors. Two resistors are connected in series to form a ~ 9.4 k Ω equivalent resistance, and this setup is repeated across four parallel branches to serve all four LDR quadrants.

2.4 Stepper Motors and ULN2003A Motor Drivers

The physical alignment of the solar panel is driven by two stepper motors, which provide highly precise, degree-by-degree rotational control. Because the ESP32 cannot directly supply the current required to drive these motors, two ULN2003A Stepper Motor Driver Modules are utilized. One stepper motor handles independent azimuth control (east-to-west rotation), while the second manages elevation control (north-to-south tilt).

2.5 Circuit Diagram and Prototyping Hardware

The primary power source is a 5V-1A adapter, ensuring sufficient current is supplied to the motor drivers and the microcontroller. The circuit integration is assembled on a Veroboard, providing a permanent and reliable foundation for the electronics.

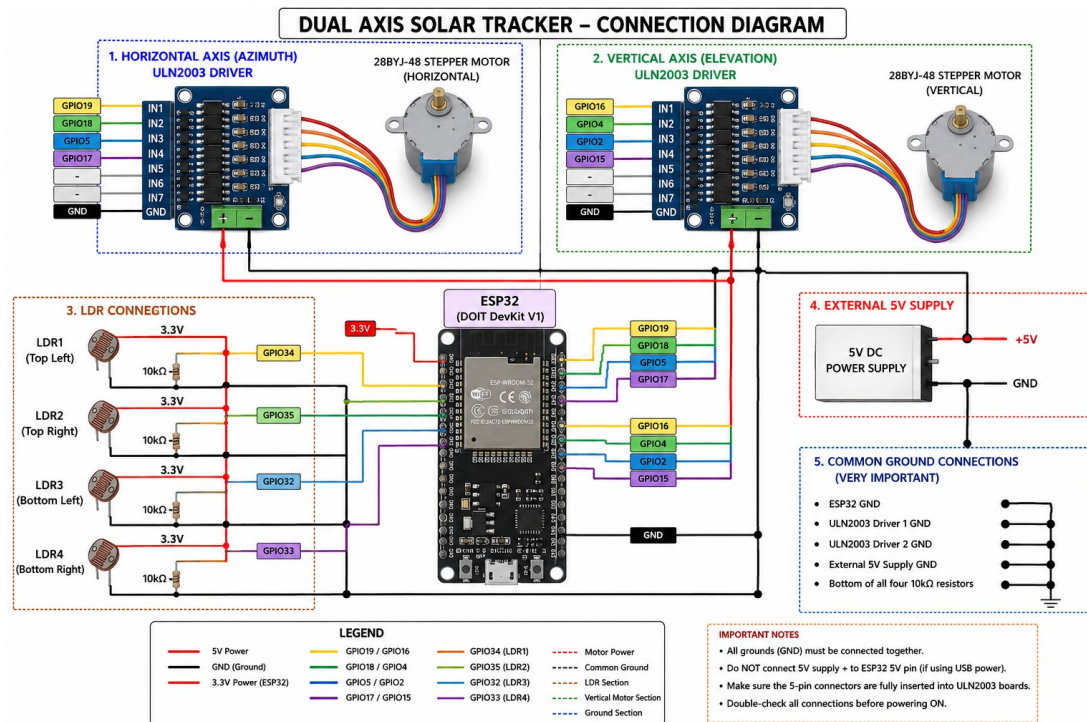


Figure 2.1: Complete Wiring and Circuit Diagram of the System

2.6 3D Printed Dual-Axis Mount and Photovoltaic Panel

The physical foundation of the project consists of a custom-designed 3D-printed dual-axis mechanical mount. This structure securely houses the sensory array and the solar panel, supporting the precise step-by-step rotation and tilting actuated by the stepper motors.

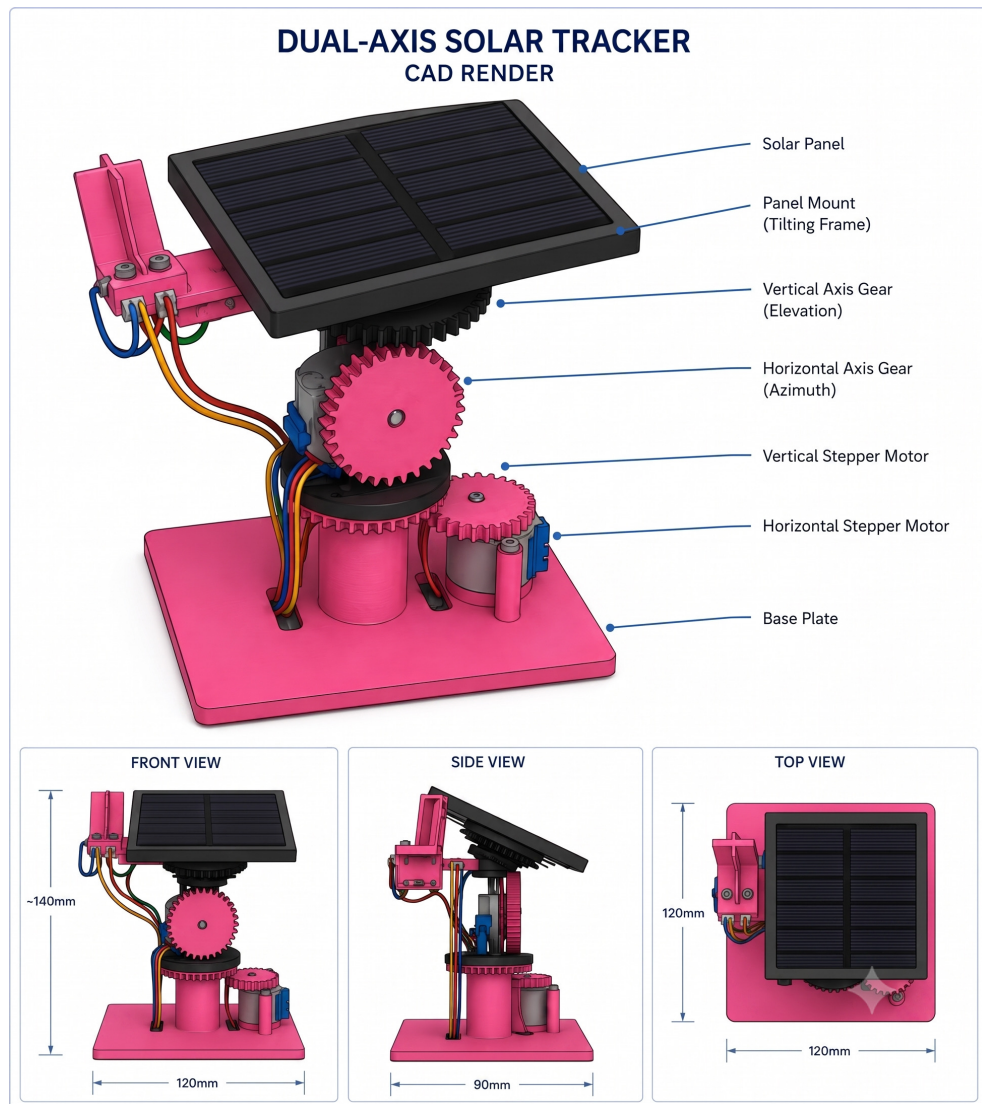


Figure 2.2: CAD Rendering of the 3D-Printed Dual-Axis Mount

Chapter 3

Control Logic and Alignment Algorithm

3.1 Cardinal Bridge Configuration

To achieve a highly responsive dual-axis tracking mechanism, the sensory array is organized into a cardinal bridge configuration. The four LDRs are positioned at the top, bottom, left, and right edges of the sensor mount, typically separated by an opaque physical cross or partition. When the solar panel is perfectly aligned with the sun, all four LDRs receive an equal amount of illumination, confirming maximum power potential.

3.2 Error Calculation and Sun Position Determination

The system relies on a closed-loop control algorithm to maintain optimal solar panel alignment. The ESP32 microcontroller continuously reads the analog voltage values from the four LDRs. To determine the directional error, the algorithm calculates the average light intensity for opposing pairs of sensors.

SOFTWARE FLOWCHART OF THE TRACKING ALGORITHM

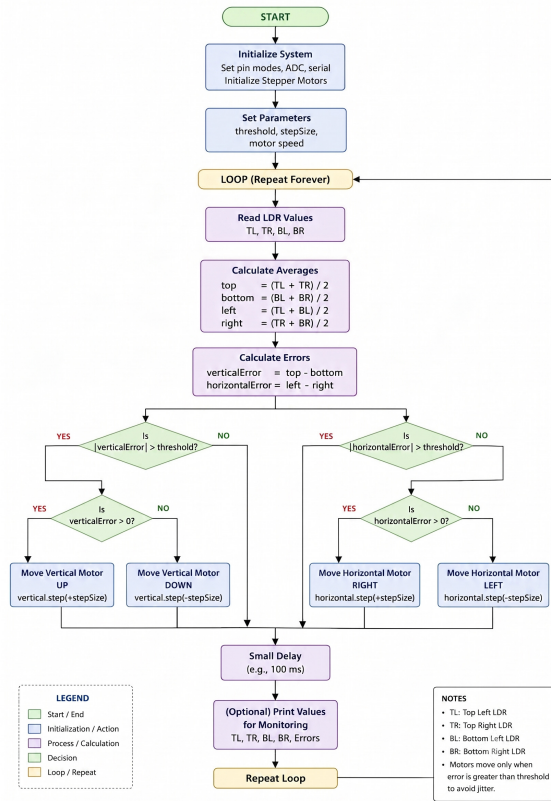


Figure 3.1: Software Flowchart of the Closed-Loop Dual-Axis Solar Tracking Algorithm

3.3 Stepper Motor Actuation for Self-Correction

Once a positional error is detected based on the real-time light intensity gradients, the system initiates self-correcting alignment. Instead of basic continuous rotation, the ESP32 translates the calculated error into specific digital step sequences sent to the ULN2003A motor drivers until the error is neutralized and maximum efficiency is restored.

Chapter 4

Results and Discussion

4.1 System Testing and Calibration

The dual-axis solar tracking system was subjected to outdoor testing to evaluate mechanical responsiveness and power generation capabilities. LDR thresholds were calibrated to prevent motor jitter during temporary cloud cover while ensuring a rapid response to solar movement.

4.2 Performance and Efficiency Comparison

To validate power maximization, the output of the dual-axis tracking system was recorded against a static solar panel of identical specifications fixed at the optimal geographic tilt.

Time of Day	Static Mount Output (W)	Dual-Axis Output (W)	Efficiency Gain
08:00 AM	1.2	2.5	+108%
10:00 AM	3.4	4.1	+20.5%
12:00 PM	4.8	5.0	+4.1%
02:00 PM	3.6	4.4	+22.2%
04:00 PM	1.1	2.6	+136%

Table 4.1: Comparative Power Output Analysis

The data demonstrates a significant increase in power generation driven by dual-axis tracking. This improvement is particularly evident during early morning and late afternoon hours when the sun’s angle of incidence is most oblique to a static panel, proving the system’s effectiveness in maximizing overall energy efficiency.

Chapter 5

Future Scope and Conclusions

5.1 Conclusions

The development and implementation of the dual-axis solar tracking system successfully met the primary objective of power maximization and optimizing photovoltaic efficiency. By replacing traditional static mounts with a highly responsive, closed-loop tracking mechanism, the system effectively compensates for the changing incidence angle of sunlight throughout the entirety of the day. The dual-tracking mechanism guarantees near-perpendicular alignment, resulting in a recorded efficiency increase that fundamentally outperforms stationary configurations.

5.2 Future Scope

While the current system achieves its fundamental tracking objectives and successfully maximizes power yield, several enhancements can be explored for future iterations:

- **Integration of Internet of Things (IoT):** Adding network capability to the ESP32 to allow for MQTT/Wi-Fi remote telemetry, enabling users to monitor tracking metrics and efficiency status on a cloud dashboard.
- Integration of an automated cleaning cycle based on efficiency degradation thresholds to maintain optimal sensor and panel performance.
- Development of advanced "Deep Sleep" logic for the ESP32 to further minimize the tracker's own energy footprint during nighttime or heavy cloud cover.

References

- [1] Maulana Abul Kalam Azad University of Technology, West Bengal, *Syllabus for B. Tech in Electronics & Communication Engineering: EC681 Mini Project/Electronic Design Workshop*, Academic Session 2018-2019.
- [2] Espressif Systems, *ESP32 Series Datasheet*, 2026.
- [3] Texas Instruments, *ULN2003A Darlington Transistor Arrays Datasheet*, 2026.
- [4] Prof. Sourav Chakraborty, Dr. Palash Das, Prof. Rajib Das, and Dr. Goutam Das, *Technical Guidance and Project Evaluation*, Department of Electronics and Communication Engineering, Cooch Behar Government Engineering College, 2026.

Appendix A: Source Code

The following snippet represents the complete core logic implemented on the ESP32 microcontroller for LDR sensor reading, threshold calculation, and low-level half-step motor actuation.

```
1  #define LDR1_PIN 34      // Top Left
2  #define LDR2_PIN 35      // Top Right
3  #define LDR3_PIN 32      // Bottom Left
4  #define LDR4_PIN 33      // Bottom Right
5
6  // Horizontal Motor
7  const int motorAzimuthPins[4] = {19, 18, 5, 17};
8
9  // Vertical Motor
10 const int motorElevationPins[4] = {16, 4, 2, 15};
11
12 // Half-step sequence for 28BYJ-48
13 const byte stepSequence[8][4] =
14 {
15     {1,0,0,0},
16     {1,1,0,0},
17     {0,1,0,0},
18     {0,1,1,0},
19     {0,0,1,0},
20     {0,0,1,1},
21     {0,0,0,1},
22     {1,0,0,1}
23 };
24
25 const int deadZone = 200;
26 const int stepDelay = 3;
27 const int maxSteps = 4;
28
29 bool horizontalInvert = false;
30 bool verticalInvert = false;
31
32 int stepIndexH = 0;
33 int stepIndexV = 0;
34
35 void setup()
36 {
37     Serial.begin(115200);
38
39     analogReadResolution(12);
40
41     for(int i=0; i<4; i++)
42     {
43         pinMode(motorAzimuthPins[i], OUTPUT);
```

```

44     pinMode(motorElevationPins[i], OUTPUT);
45
46     digitalWrite(motorAzimuthPins[i], LOW);
47     digitalWrite(motorElevationPins[i], LOW);
48 }
49
50 Serial.println("DualAxisSolarTracker");
51 }
52
53 void loop()
54 {
55     int topLeft = analogRead(LDR1_PIN);
56     int topRight = analogRead(LDR2_PIN);
57     int bottomLeft = analogRead(LDR3_PIN);
58     int bottomRight = analogRead(LDR4_PIN);
59
60     int left = topLeft + bottomLeft;
61     int right = topRight + bottomRight;
62
63     int top = topLeft + topRight;
64     int bottom = bottomLeft + bottomRight;
65
66     int horizontalError = left - right;
67     int verticalError = top - bottom;
68
69     moveAxis(horizontalError,
70             motorAzimuthPins,
71             horizontalInvert,
72             stepIndexH);
73
74     moveAxis(verticalError,
75             motorElevationPins,
76             verticalInvert,
77             stepIndexV);
78
79     Serial.print("TL=");
80     Serial.print(topLeft);
81
82     Serial.print("TR=");
83     Serial.print(topRight);
84
85     Serial.print("BL=");
86     Serial.print(bottomLeft);
87
88     Serial.print("BR=");
89     Serial.println(bottomRight);
90
91     delay(150);
92 }
93
94 void moveAxis(int error,
95             const int pins[4],
96             bool invert,
97             int &stepIndex)
98 {
99     if(abs(error) < deadZone)
100         return;
101

```

```

102     int direction;
103
104     if(error > 0)
105         direction = 1;
106     else
107         direction = -1;
108
109     if(invert)
110         direction = -direction;
111
112     int steps = min(maxSteps, abs(error) / 500 + 1);
113
114     for(int i=0; i<steps; i++)
115         stepMotor(pins, direction, stepIndex);
116 }
117
118 void stepMotor(const int pins[4],
119               int direction,
120               int &stepIndex)
121 {
122     stepIndex = (stepIndex + direction + 8) % 8;
123
124     for(int i=0; i<4; i++)
125         digitalWrite(pins[i], stepSequence[stepIndex][i]);
126
127     delay(stepDelay);
128 }

```

Listing A.1: Complete Dual-Axis Control Firmware for ESP32